Project Report: HR Attrition Prediction (Enterprise Simulation)

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1. Data Understanding Report

1.1. Problem Statement

The organization is experiencing a noticeable increase in employee attrition, which is leading to:

1. Increased hiring and onboarding costs
2. Loss of experienced workforce
3. Reduced productivity and team stability
4. Impact on overall business performance

Business Context

Organizations face significant challenges due to increasing employee attrition.

A predictive system can:

- Reduce hiring and training costs
- Improve employee retention strategies
- Identify high-risk employees in advance
- Enhance workforce planning and productivity
- Maintain team stability and business performance

1.2. Objective

The objective is to build a machine learning solution to predict employee attrition (Yes/No) and identify key factors driving it.

The key objectives of this project are:

- Develop a predictive model to identify employees likely to leave the organization
- Understand key factors contributing to employee attrition
- Enable HR teams to take preventive actions
- Reduce overall attrition rate and associated costs

1.3. Target Variable

Attrition
1→Yes
0→No

1.4. Input Variable

Typical features include:

Age	EnvironmentSatisfaction	MonthlyIncome	StockOptionLevel
BusinessTravel	Gender	MonthlyRate	TotalWorkingYears
DailyRate	HourlyRate	NumCompaniesWorked	TrainingTimesLastYear
Department	JobInvolvement	OverTime	WorkLifeBalance
DistanceFromHome	JobLevel	PercentSalaryHike	YearsAtCompany
Education	JobRole	PerformanceRating	YearsInCurrentRole
EducationField	JobSatisfaction	RelationshipSatisfaction	YearsSinceLastPromotion
EmployeeNumber	MaritalStatus	StandardHours	YearsWithCurrManager

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1.5. Initial Data Checks

Performed:

- df.shape → Dataset dimensions
- df.info() → Data types and null values
- df.describe() → Statistical summary
- df.head() / df.tail() → Sample inspection

Observations

- We have total 1470 Rows and 35 column in the data set
- There are no null values in dataset

1.6. Data Cleaning Summary

Unique Columns Removed

- Columns with all unique values (Over18 & EmployeeCount)

Null Value Handling:

There are no null values present in the dataset

Column Type	Strategy
Numerical (float)	Mean Imputation
Numerical (int)	Median Imputation
Categorical	Mode Imputatio

Duplicate Handling

- Checked using df.duplicated().sum()
- Removed using df.drop_duplicates()

Error Value Handling

- Replaced values like "?", "unknown" with NaN
- Treated as missing values

Column Segregation

- Categorical columns
- Numerical columns

1.7. Outlier Detection & Treatment

Technique Used: IQR Method

- Lower Bound = Q1 - 1.5 * IQR
- Upper Bound = Q3 + 1.5 * IQR

Action

- Removed extreme outliers beyond bounds

1.8. Exploratory Analysis Summary

Univariate Analysis

- Distribution of EnvironmentSatisfaction, NumCompaniesWorked, Age, JobSatisfaction, YearsAtCompany, YearsInCurrentRole, PercentSalaryHike, MonthlyIncome
- Frequency of Attrition

Bivariate Analysis

- Heat Map

Key Observations

- More employees staying than leaving.
- Most employees fall in the 25–40 age group.
- Income distribution is right-skewed (most employees earn lower to mid-range salaries).
- Fewer employees are in higher salary brackets, which may influence retention.
- Majority of employees have less than 10 years of experience at the company.
- Higher attrition is often seen in early tenure (0–4 years).
- Most employees have spent 2–4 years in the same role.
- Limited role changes may indicate stagnation, potentially leading to attrition.

- Many employees have worked in 1–3 companies, indicating moderate job switching behavior.
- Most employees receive moderate salary hikes (11–15%).
- Lack of higher increments could be a factor in employee dissatisfaction.

2. Feature Engineering Documentation

2.1. Categorical Encoding

Techniques Used: Label Encoding

Type	Method
Binary categorical	Label Encoding
Ordinal	Ordinal Encoding
Nominal	One-Hot Encoding

2.2 Feature Transformation

- Log transformation applied (if skewed)
- Scaling applied where required (for linear models)

2.3 Feature Selection

Removed:

- Non-informative features (Over18 & EmployeeCount)
- Highly correlated features (if multicollinearity detected)

2.4 Final Feature Set

- Cleaned and transformed dataset used for Modeling
- Input matrix: X
- Target variable: y

3. Model Development and Evaluation Report

3.1 Dataset Splitting

- Train-Test Split:
- 70% training
- 30% testing

Outputs:

- X_train, X_test

- y_train, y_test

3.2 Models Implemented

1. Logistic Regression

- Baseline model
- Interpretable

2. Decision Tree

- Handles non-linearity
- Easy to visualize

3. Random Forest

- Ensemble model
- Reduces overfitting

3.3 Model Training

- Models trained using:
- model.fit(X_train, y_train)

3.4 Model Evaluation Metrics Used (Classification)

- Confusion Matrix
- Accuracy
- Precision
- Recall
- Misclassification Rate

3.5 Results Summary

Model	Accuracy	Precision	Recall
Logistic Regression	0.83	0.75	0.076
Naive Bayes	0.75	0.37	0.58
Random Forest	0.86	0.75	0.15
Decision Tree	0.75	0.34	0.41

3 .6 Model Selection

- Selected Model: Random Forest

Reason

- Highest accuracy

- Better generalization
- Handles feature interactions well

4. Insights and Recommendations Report

4.1 Key Insights

- OverTime is the most important predictor of attrition
- Employees who are single tend to leave more frequently
- Sales roles show higher attrition rates
- Employees who travel frequently are more likely to leave
- Higher salary and job level reduce attrition risk
- Longer tenure with company and manager improves retention

4.2 Feature Importance (Example)

Top contributing features:

- OverTime
- MaritalStatus (Single)
- JobRole (Sales Representative)
- BusinessTravel (Frequent Travel)
- MonthlyIncome
- JobLevel

4.3 Business Recommendations

1. Reduce Employee Overload

- Monitor and limit overtime
- Promote work-life balance initiatives

2. Improve Compensation Strategy

- Offer competitive salaries
- Introduce performance-based incentives

3. Target High-Risk Roles

- Focus on Sales Department and frequently traveling employees
- Provide additional support and incentives

4. Enhance Employee Engagement

- Improve job satisfaction programs
- Conduct regular feedback sessions

4.4 Model Usage

- Input: Employee details (salary, job role, overtime, etc.)
- Output: Attrition prediction (0 = Stay, 1 = Leave)

4.5 Future Enhancements

- Use advanced models (Random Forest, XGBoost)
- Include employee feedback/survey data
- Deploy model via API
- Build dashboard for real-time attrition monitoring

5. Conclusion

The project successfully analysed employee data to identify key factors influencing attrition. It was observed that overtime, job role, business travel, and employee demographics significantly impact attrition rates, while salary, job level, and tenure contribute to employee retention.

The insights derived from the analysis can help organizations take proactive measures to improve employee satisfaction, reduce turnover, and enhance workforce stability. Overall, this study provides valuable guidance for HR decision-making and strategic planning.